Statistics & Econometrics 2025

Tutorial 3: Problem Set

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1 Time-series Econometrics

1. Assume your firm manages a government bond portfolio that tracks the slope of the U.S. Treasury curve, specifically the difference between the ten-year Treasury and the three-month Treasury .

   You obtain the following result:

   $$y_t = 0.1 + 0.95 y_{t-1} + \hat{u}_t$$

   where the slope y_t is sampled at monthly frequency and where the variance of the innovation $\hat{u}_t$ is $Var(\hat{u}_t) = 0.16$

   Question: To better understand the dynamics and long-run properties of the resulting portfolio, your manager asks you to compute its implied long-run mean and variance.

   What are the unconditional mean and variance of the estimated AR(1) process for the Treasury slope?

   Solution: The unconditional mean of an AR(1) process

   $$y_t = \mu + \phi_1 y_{t-1} + u_t, \qquad u_t \sim N(0, \sigma_u^2)$$

   is given by

   $$E[y] = \frac{\mu}{1 - \phi_1}$$

   and the unconditional variance by

   $$Var(y) = \frac{\sigma_u^2}{1 - \phi_1^2}$$

   This can be shown by recursive substitution or in the following way:

   $$\begin{aligned} y_t &= \mu + \phi_1 y_{t-1} + u_t \\ E[y_t] &= E[\mu + \phi_1 y_{t-1} + u_t] \\ &= \mu + \phi_1 E[y_{t-1}] + 0 \end{aligned}$$

   As $|\phi_1| < 1$, the long-run expected value of $E[y_t]$ and $E[y_{t-1}]$ are the same (they are forecasts of y_t and y_{t-1} infinitely many periods into the future), hence

$$\begin{aligned} E[y_t] &= \mu + \phi_1 E[y_t] \\ (1-\phi_1)E[y_t] &= \mu \\ E[y_t] = \frac{\mu}{1-\phi_1} & \end{aligned}$$

In a similar vein, we have

$$\begin{aligned} y_t &= \mu + \phi_1 y_{t-1} + u_t \\ Var(y_t) &= Var(\mu + \phi_1 y_{t-1} + u_t) \\ &= Var(\mu) + \phi_1^2 Var(y_{t-1}) + Var(u_t) \\ &= 0 + \phi_1^2 Var(y_{t-1}) + \sigma_u^2 \end{aligned}$$

But the unconditional variance of y_t and y_{t-1} are the same, so as above

$$\begin{aligned} Var(y_t) &= \phi_1^2 Var(y_t) + \sigma_u^2 \\ (1-\phi_1^2)Var(y_t) &= \sigma_u^2 \\ Var(y_t) &= \frac{\sigma_u^2}{1-\phi_1^2} \end{aligned}$$

Applied to the process above, we therefore find

$$E[y_t] = \frac{0.1}{1-0.95} = 2$$

and the unconditional variance by

$$Var(y_t) = \frac{0.16}{1-0.95^2} = 1.64$$

2. Your manager finds these values plausible and next asks you to use the model to predict the Treasury slope for the next period, two periods out and three periods out. The current level of the 3-month Treasury Bill is 5.3 and the current level of the ten-year Treasury note is 4.2.

   Question: What are the predicted values of the Treasury slope for the next three months?

   Solution: To predict this process based on currently available information, we need to compute the *conditional* mean of this AR(1) process. This is given by

$$\begin{aligned} E_t[s_{t+1}] &= E_t[\mu + \phi_1 s_t + u_{t+1}] \\ &= \mu + \phi_1 s_t \\ &= 0.1 + 0.95 \cdot (4.2 - 5.3) \\ &\approx -0.95 \end{aligned}$$

The conditional two-step ahead expectation $E_t[s_{t+2}]$ is given by

$$\begin{aligned} E_t[s_{t+2}] &= E_t[\mu + \phi_1 s_{t+1} + u_{t+2}] \\ &= \mu + \phi_1 E_t[s_{t+1}] \\ &= \mu + \phi_1(\mu + \phi_1 s_t) \\ &= \mu + \phi_1 \mu + \phi_1^2 s_t \\ &= 0.1 + 0.95 \cdot 0.1 + 0.95^2 \cdot (-1.1) \\ &\approx -0.80 \end{aligned}$$

Similarly, The conditional three-step ahead expectation $E_t[s_{t+3}]$ is given by

$$\begin{aligned} E_t[s_{t+3}] &= E_t[\mu + \phi_1 s_{t+2} + u_{t+3}] \\ &= \mu + \phi_1 E_t[s_{t+2}] \\ &= \mu + \phi_1(\mu + \phi_1 \mu + \phi_1^2 s_t) \\ &= \mu(1 + \phi_1 + \phi_1^2) + \phi_1^3 s_t \\ &= 0.1 \cdot (1 + 0.95 + 0.95^2) + 0.95^3 \cdot (-1.1) \\ &\approx -0.66 \end{aligned}$$

3. Your manager is surprised by these results. He/She would have expected the Treasury slope to return to zero more quickly. He/She therefore asks you to also compute the one-year ahead ahead forecast.

   Question: What is the one-year ahead forecast of the Treasury slope implied by the estimated model?

   Solution: The above conditional forecasts for $h = 1, 2, 3$ months into the future can be generalized in the following way:

$$\begin{aligned} E_t[s_{t+h}] &= E_t[\mu + \phi_1 s_{t+h-1} + u_{t+h}] \\ &= \mu(1 + \phi_1 + \phi_1^2 + \ldots + \phi_1^{h-1}) + \phi_1^h s_t \end{aligned}$$

Denote the sum $1 + \phi_1 + \phi_1^2 + \ldots + \phi_1^{h-1}$ by S_h. You can easily compute this sum in the following simple way

$$\begin{aligned}
S_h &= 1 + \phi_1 + \phi_1^2 + \ldots + \phi_1^{h-1} \\
(1-\phi_1)S_h &= (1-\phi_1)(1 + \phi_1 + \phi_1^2 + \ldots + \phi_1^{h-1}) \\
&= (1 - \phi_1 + \phi_1 - \phi_1^2 + \phi_1^2 - \phi_1^3 + \ldots + \phi_1^{h-1} - \phi_1^{h-1} - \phi_1^h) \\
&= (1 - \phi_1^h)
\end{aligned}$$

Therefore,

$$S_h = \frac{1-\phi_1^h}{1-\phi_1}$$

and thus the $h-$step ahead forecast of s_t is given by

$$E_t[s_{t+h}] = \frac{\mu(1-\phi_1^h)}{1-\phi_1} + \phi_1^h s_t$$

Hence, the one-year ahead forecast is given by

$$\begin{aligned}
E_t[s_{t+12}] &= \frac{\mu(1-\phi_1^{12})}{1-\phi_1} + \phi_1^{12} s_t \\
E_t[s_{t+12}] &= \frac{0.1(1-0.95^{12})}{1-0.95} + 0.95^{12} \cdot (-1.1) \\
&\approx 0.33
\end{aligned}$$

4. This appears more plausible to your manager. He now asks you to analyze what the effect of a surprise hike of the federal funds rate by 25 basis points would imply for the dynamics of the Treasury slope, assuming that the Fed's move would lead to an immediate increase in the three-month Treasury Bill by 25 bps but leave the ten-year Treasury yield unaffected. Specifically, he asks you the following.

Question: What is the effect of such a shock $u_t = 0.25$ on s_{t+6}, i.e. the Treasury slope in six months time?

Solution: One can write

$$\begin{aligned}
s_t &= \mu + \phi_1 s_{t-1} + u_t \\
&= \mu + \phi_1(\mu + \phi_1 s_{t-2} + u_{t-1}) + u_t \\
&= \mu + \phi_1\mu + \phi_1^2 s_{t-2} + \phi_1 u_{t-1} + u_t \\
&= \mu + \phi_1\mu + u_t + \phi_1 u_{t-1} + \phi_1^2(\mu + \phi_1 s_{t-3} + u_{t-2}) \\
&= \sum_{i=0}^{t} \mu\phi_1^i + \sum_{i=0}^{t} \phi_1^i u_{t-i} + \phi_1^t y_0 \ldots
\end{aligned}$$

Hence, a shock $u_{t-6} = 0.25$ would have increased s_t by $\phi_1^6 \cdot 0.25$. Similarly, a shock $u_t = 0.25$ would increase s_{t+6} by $\phi_1^6 \cdot 0.25 = 0.18$ percentage points or 18 basis points.

One can also see this by comparing the conditional forecast with the shock hitting with the counterfactual forecast with the shock not hitting s_t. Denote the former $E_t[s_{t+6}]$:

$$\begin{aligned} E_t[s_{t+6}] &= \frac{\mu(1-\phi_1^6)}{1-\phi_1} + \phi_1^6 s_t \\ &= \frac{\mu(1-\phi_1^6)}{1-\phi_1} + \phi_1^6(\mu + \phi_1 s_{t-1} + u_t) \end{aligned}$$

Denote the latter $E_t[\tilde{s}_{t+6}]$:

$$\begin{aligned} E_t[\tilde{s}_{t+6}] &= \frac{\mu(1-\phi_1^6)}{1-\phi_1} + \phi_1^6 \tilde{s}_t \\ &= \frac{\mu(1-\phi_1^6)}{1-\phi_1} + \phi_1^6(\mu + \phi_1 s_{t-1} + 0) \end{aligned}$$

Then, the difference between the two conditional forecasts is given by

$$E_t[s_{t+6}] - E_t[\tilde{s}_{t+6}] = \phi_1^6 \cdot u_t = 0.95^6 \cdot 0.25 = 0.18.$$

5. Your manager is quite happy with your analytical skills and puts you in charge of evaluating the risk-return tradeoff for a potential investment in a hedge fund. You are provided with 15 years of monthly data of the fund's past returns, expressed in percentage points. Note: Since returns are in percentage points, all calculations preserve these units—the variance will be in ($\%^2$) and standard deviation in (%).

   You estimate the following AR(1) - ARCH(1) model for these returns:

$$\begin{aligned} r_t &= 1\% + 0.15 r_{t-1} + \hat{u}_t, \qquad u_t \sim N(0, \sigma_t^2) \\ \sigma_t^2 &= 7\%^2 + 0.9\hat{u}_{t-1}^2 \end{aligned}$$

   Assume further that the current (period-t) risk-free rate is given by $r_{f,t} = 0.2\%$ and that the last two observations of r are given by $r_t = 2\%$ and $r_{t-1} = -2\%$, respectively.

   <u>Question:</u> Compute the conditional Sharpe ratio $SR_t = \frac{E_t[r_{t+1}] - r_{f,t}}{\sqrt{Var_t(r_{t+1})}}$

Solution: The conditional expected return is given by

$$E_t[r_{t+1}] = 1\% + 0.15 \cdot 2\% = 1.3\%$$

Hence, the expected one-period ahead *excess* return is given by

$$E_t[r_{t+1}] - r_{f,t} = 1.3\% - 0.2\% = 1.1\%$$

The conditional variance of the expected return is given by

$$\begin{aligned}
Var_t(r_{t+1}) &= Var_t(\hat{u}_{t+1}) = \sigma^2_{t+1} \\
&= 7\%^2 + 0.9 \cdot \hat{u}_t^2 \\
&= 7\%^2 + 0.9 \cdot (r_t - 1\% - 0.15 \cdot r_{t-1})^2 \\
&= 7\%^2 + 0.9 \cdot (2\% - 1\% - 0.15 \cdot (-2\%))^2 \\
&= 7\%^2 + 1.521\%^2 \\
&= 8.52\%^2
\end{aligned}$$

Hence, the conditional Sharpe ratio is given by

$$SR_t = \frac{E_t[r_{t+1}] - r_{f,t}}{\sqrt{Var_t(r_{t+1})}} = \frac{1.1\%}{2.9191\%} = 0.3768$$

2 Panel Regressions

You work in a bank and observe the total loan volume L_{it} the bank has to each firm i. You want to know if the firm size is an important explanation for your loan demand.

$$L_{it} = \beta_0 + \beta_1 \text{Assets}_{it} + \epsilon_{it}$$

1. Your manager is worried that macroeconomic factors drive variation in both loan demand and size of the firms. What type of specification would alleviate your manager's concern?
   Solution: You should estimate a model with a time-fixed effect.

   $$L_{it} = \beta_0 + \beta_1 \text{Assets}_{it} + \beta_2 S_t + \epsilon_{it}$$

   This can be implemented with a regression that includes a time-fixed effect. The time-fixed effect absorbs the effects of all variables that vary over t and affect $L_i t$ linearly.

2. Your manager additionally issues the concern that the firm's headquarter location can also explain their demand for loans. (Suppose firm's never change their

headquarter location). What type of specification would alleviate your manager's concern?
Solution: You should additionally estimate a model with firm fixed-effect.

$$L_{it} = \beta_0 + \beta_1 \text{Assets}_{it} + \beta_2 S_t + \beta_3 Z_i + \epsilon_{it}$$

The firm-fixed effect absorbs the effects of all variables that vary over i and affect $L_i t$ linearly.

3. Your manager asks you to include the firms rating into the analysis.

$$L_{it} = \beta_0 + \beta_1 \text{Assets}_{it} + \gamma \text{Rating}_{it} + \beta_2 S_t + \beta_3 Z_i + \epsilon_{it}$$

Are you able to estimate the coefficient γ on the rating variable? Which set of firms will be absorbed by the firm-fixed effect?
Solution: Yes, you're able to estimate the effects if the ratings change over time. Firms that never experienced a rating change are absorbed by the firm-fixed effect. In case, all firms experience no rating changes, the rating variable and the firm-fixed effect are collinear and it is not possible the estimate γ anymore.

4. Suppose your manager asks you to also consider a specification including the year when the company was founded. Are you able to estimate a model where you include the year the company was founded as an independent variable together with time-fixed effects and firm-fixed effects?
Solution: No, the variable is absorbed by the firm-fixed effect.